# LIBR: A Moderated, Censorship-Resilient Social Network Framework

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***Abstract*—Digital forums serve as critical infrastructure for democratic discourse, yet existing platforms suffer from fundamental limitations: centralized systems enable arbitrary censorship, while decentralized alternatives lack effective moderation mechanisms. This paper introduces LIBR, a novel framework that reconciles censorship resistance with community-driven moderation through Byzantine fault-tolerant consensus. LIBR employs a three-tier architecture comprising clients, distributed database nodes, and cryptographically-verified moderators. The system achieves partial immutability through replicated distributed hash tables (DHTs) while enabling content removal via Byzantine consensus among community moderators. We formally analyze LIBR's security properties, proving safety and liveness under standard Byzantine assumptions. We also present comprehensive incentivization and governance mechanisms that ensure sustainable operation and fair resource allocation. Our theoretical analysis demonstrates LIBR achieves $O(\log n)$ message complexity for content operations with tolerance of up to $\lfloor (m-1)/2 \rfloor$ malicious moderators among $m$ total moderators.**

***Index Terms*—Byzantine fault tolerance, censorship resistance, distributed systems, peer-to-peer networks, content moderation, digital forums, distributed hash tables**

## I. Introduction

Digital forums constitute the backbone of online discourse, facilitating knowledge exchange, community building, and democratic participation. However, the dichotomy between centralized and decentralized forum architectures presents fundamental trade-offs that remain unresolved in current systems.

Centralized platforms, exemplified by Reddit, Twitter (X), and traditional bulletin boards, provide intuitive user experiences and effective moderation capabilities. Yet they suffer from critical vulnerabilities: single points of failure, susceptibility to censorship by platform operators or external authorities, and opaque moderation decisions that lack accountability. Recent events have highlighted these limitations, with platforms arbitrarily removing content based on political pressure or commercial interests [1].

Conversely, decentralized alternatives like blockchain-based forums or peer-to-peer networks offer censorship resistance through distributed architecture. However, they typically lack effective moderation mechanisms, leading to proliferation of harmful content, spam, and antisocial behavior that degrades user experience and community cohesion [2].

This fundamental tension between censorship resistance and content quality has remained a persistent challenge in designing digital communication platforms. Existing solutions either sacrifice user safety for decentralization or compromise on censorship resistance to maintain usability.

### A. Contributions

This paper presents LIBR, a novel framework that addresses this dichotomy through the following key contributions:

1) **Hybrid Architecture**: A three-tier system combining distributed storage, Byzantine fault-tolerant moderation, and client-side interfaces that preserves decentralization while enabling effective governance.
2) **Partial Immutability Model**: A novel approach to content persistence that prevents unauthorized modification while allowing community-driven removal through cryptographic consensus.
3) **Formal Security Analysis**: Rigorous proofs of safety, liveness, and censorship resistance properties under standard Byzantine assumptions.
4) **Comprehensive Governance Framework**: A novel incentivization and governance model that ensures sustainable node operation and democratic moderator selection through cryptoeconomic mechanisms.
5) **Theoretical Analysis**: Complete theoretical characterization of system properties including complexity bounds, fault tolerance guarantees, and scalability analysis.

The remainder of this paper is organized as follows: Section II surveys related work in decentralized social media and distributed systems. Section III provides necessary background and preliminaries. Section IV formalizes the problem and system model. Section V presents the LIBR architecture and protocols. Section VI describes our incentivization and governance mechanisms. Section VII provides formal security analysis. Section VIII presents theoretical analysis and complexity bounds. Section IX discusses limitations and future work, and Section X concludes.

## II. Related Work

### A. Centralized Social Media Platforms

Traditional centralized platforms dominate the current social media landscape, offering polished user experiences but introducing fundamental structural limitations.

**Reddit** represents the archetypal community-driven forum with hierarchical moderation. Communities (subreddits) operate under community-elected moderators with significant autonomy in rule enforcement. However, the platform maintains ultimate administrative control, enabling arbitrary content removal and community bans. The 2020-2021 content policy changes and community shutdowns demonstrated the vulnerability of centralized governance to external pressure and internal policy shifts [25].

**Twitter/X** exemplifies microblogging platforms with algorithmic content curation and centralized moderation policies. The platform's acquisition by Elon Musk in 2022 and subsequent policy reversals highlighted the instability inherent in corporate-controlled discourse platforms. Changes to verification systems, content moderation policies, and algorithmic feeds demonstrated how centralized control enables rapid, unilateral alterations to user experience and content visibility [26].

These platforms share common vulnerabilities: single points of failure, susceptibility to regulatory capture, opaque algorithmic manipulation, and the potential for arbitrary policy enforcement. Users have limited recourse when content is removed or accounts suspended, and entire communities can be eliminated without meaningful appeal processes.

### B. Decentralized Social Media Platforms

The landscape of decentralized social media has evolved through several paradigms, each addressing different aspects of the centralization problem.

**Blockchain-Based Platforms**: Steemit [3] and its successor Hive employ blockchain technology with economic incentives for content creation and curation. While achieving immutability, these platforms suffer from scalability limitations (3-4 TPS) and plutocratic governance where influence correlates with token holdings. Voice, built on EOS, attempted to address manipulation through identity verification but introduced centralized trust assumptions incompatible with decentralization principles.

**Federated Systems**: Mastodon implements the ActivityPub protocol [4], creating interconnected but independently operated instances. While providing administrative decentralization, the architecture maintains centralization within instances, enabling administrator-level censorship and creating single points of failure. Similarly, Diaspora's pod-based architecture faces reliability and consistency challenges across federated nodes.

**Peer-to-Peer Networks**: IPFS-based social media experiments leverage content-addressed storage for censorship resistance [5]. However, these systems lack native moderation mechanisms and struggle with content discoverability and long-term availability without additional incentive layers.

**Nostr Protocol**: Nostr (Notes and Other Stuff Transmitted by Relays) represents a novel approach to decentralized social media through cryptographically signed messages distributed across multiple relays [27]. Unlike federated systems, Nostr users maintain portable identities through cryptographic key pairs, enabling seamless migration between relays without losing social connections. The protocol's "outbox model" ensures censorship resistance by allowing users to publish to multiple relays simultaneously. While Nostr achieves strong censorship resistance and user sovereignty, it faces challenges in content moderation, as each relay independently decides what content to store and serve, potentially leading to inconsistent content availability across the network.

### C. Byzantine Fault Tolerance in Distributed Systems

Byzantine fault tolerance forms the theoretical foundation for our moderation mechanism. The seminal work by Lamport et al. [6] established fundamental limits for Byzantine agreement, proving impossibility with $f \geq n/3$ Byzantine nodes among $n$ total nodes.

Practical Byzantine Fault Tolerance (PBFT) [7] demonstrated feasible Byzantine consensus requiring $3f + 1$ nodes to tolerate $f$ failures. Subsequent work has explored various optimizations and extensions, including BFT-SMaRt [8] for state machine replication and HotStuff [9] for improved performance.

Our work extends these foundations by applying Byzantine consensus specifically to content moderation, introducing simple majority voting mechanisms while preserving formal security guarantees.

### D. Distributed Hash Tables

DHT systems like Chord [10] and Kademlia [11] provide decentralized key-value storage with logarithmic lookup complexity. LIBR utilizes a custom implementation of Kademlia DHT as the backbone of content storage while preserving fault tolerance.

Recent work has explored DHT applications in social media contexts, including Twister [12] which combines DHT storage with blockchain user registration. However, these systems lack coherent moderation frameworks, thus not utilized for our case.

### E. Content Moderation in Decentralized Systems

Traditional content moderation relies on centralized authorities with opaque decision-making processes. Recent research has explored algorithmic approaches [13] and community-driven moderation [14], but these remain primarily centralized implementations.

Blockchain-based governance systems like Aragon [15] and decentralized autonomous organizations (DAOs) provide frameworks for community decision-making but are typically designed for organizational governance rather than real-time content moderation.

**Recent Advances in Decentralized Moderation**: The Fediverse has emerged as a testbed for collaborative content

moderation approaches. Zia et al. [16] introduced FedMod, a federated learning-based system that enables servers to exchange partially trained moderation models, achieving robust performance across harmful content detection, bot detection, and content warning assignment. Their work demonstrates the feasibility of collaborative moderation without centralized infrastructure.

Complementing this, research on decentralized blocklist mechanisms [17] has shown how community-level content filtering can be implemented across federated networks, providing users with agency over their content experience while maintaining decentralized governance.

The practical deployment of decentralized social media has provided valuable insights into real-world moderation challenges. Studies of Bluesky's AT Protocol [18] demonstrate how technical architecture can be designed to support multiple interoperable moderation providers, enabling users to choose their preferred moderation policies while maintaining system-wide interoperability.

### *F. Modern Approaches to Decentralized Social Networks*

Recent developments in decentralized social media platforms have addressed many limitations of earlier systems through novel architectural approaches and governance mechanisms.

**AT Protocol and Bluesky**: The AT Protocol represents a significant advancement in decentralized social media architecture [18]. Unlike traditional federated systems, AT Protocol separates identity, data hosting, and algorithmic curation into distinct, interoperable layers. This separation enables users to switch providers without losing their social graph or content history, addressing a major limitation of earlier federated systems. The protocol's approach to content moderation allows multiple moderation services to operate on the same data layer, providing users with choice while maintaining coherent community standards.

**Fediverse Evolution**: The Fediverse has evolved significantly since the early days of Diaspora and GNU Social. Recent studies of cross-platform interaction dynamics between Threads and Mastodon [19] reveal new patterns of content sharing and community formation across federated instances. This research highlights both the opportunities and challenges of maintaining community cohesion across decentralized platforms with different governance models.

Analysis of toxicity and moderation in the decentralized web [20] has shown that while decentralized platforms face unique challenges in content moderation, collaborative approaches can achieve effectiveness comparable to centralized systems when properly implemented.

### *G. Recent Advances in Byzantine Fault Tolerance*

The field of Byzantine fault tolerance has seen significant theoretical and practical advances relevant to decentralized content moderation systems.

**Scalability Improvements**: Recent BFT protocols have focused on improving scalability and reducing communication complexity. Arma [21] achieves linear scalability across all hardware resources by separating transaction dissemination from validation. Banyan [22] demonstrates that rotating leader protocols can achieve single round-trip confirmation times in Byzantine settings, significantly improving latency.

**Practical Deployments**: Modern BFT systems have explored deployment in resource-constrained environments. Research on Byzantine-robust federated learning [23] shows how BFT principles can be applied to collaborative machine learning, relevant for our approach to distributed content moderation.

**Fault Models and Resilience**: Recent work has expanded Byzantine fault models to address modern attack vectors. BigDipper [24] introduces short-term censorship resistance in BFT systems, addressing attacks where adversaries temporarily control message dissemination. These advances inform LIBR's approach to handling sophisticated adversaries who may attempt to manipulate content visibility.

LIBR bridges this gap by providing a practical framework for community-driven content moderation with formal security guarantees and real-time operation capabilities, incorporating lessons learned from recent advances in both decentralized social media and Byzantine fault tolerance.

## III. Background and Preliminaries

### *A. Cryptographic Primitives*

LIBR relies on standard cryptographic primitives that provide the security foundation for the system:

**Digital Signatures**: We use an unforgeable digital signature scheme $(KeyGen, Sign, Verify)$ where:

- $KeyGen() \rightarrow (sk, pk)$ generates a signing key and verification key pair
- $Sign(sk, m) \rightarrow \sigma$ produces a signature on message $m$
- $Verify(pk, m, \sigma) \rightarrow \{0, 1\}$ verifies signature validity

**Cryptographic Hash Functions**: We employ collision-resistant hash functions $H : \{0,1\}^* \rightarrow \{0,1\}^k$ for content addressing and integrity verification.

### *B. Distributed Hash Tables*

LIBR utilizes a Kademlia-based DHT for content storage with the following properties:

**Key Space**: The system uses a 160-bit key space with XOR distance metric for routing decisions.

**Routing Table**: Each node maintains a routing table with $O(\log n)$ entries, enabling efficient lookup in $O(\log n)$ hops.

**Replication**: Content is replicated across multiple nodes to ensure availability and fault tolerance.

### *C. Byzantine Fault Tolerance Model*

LIBR's moderation system operates under the Byzantine fault model:

**Fault Assumptions**: Up to $f$ out of $n$ nodes may exhibit arbitrary (Byzantine) behavior, including sending inconsistent messages or remaining silent.

**Synchrony Model**: We assume partial synchrony where there exists a Global Stabilization Time (GST) after which

message delivery is timely, though GST is unknown to participants.

**Cryptographic Assumptions**: Digital signatures cannot be forged by computationally bounded adversaries, and hash functions are collision-resistant.

## IV. Problem Formulation and System Model

### A. Problem Statement

We address the fundamental challenge of designing a digital forum that simultaneously satisfies the following requirements:

1) **Censorship Resistance**: No single entity can arbitrarily remove or modify content across network.
2) **Content Quality**: Mechanisms exist to remove harmful or inappropriate content through community consensus.
3) **Scalability**: The system supports large-scale deployment with efficient resource utilization.
4) **Availability**: Content remains accessible despite node failures or network partitions.
5) **Integrity**: Content authenticity and authorship can be cryptographically verified.

### B. System Model

*1) Network Model:* We assume a partially synchronous network model where:

- Message delivery is eventually reliable but may experience arbitrary delays
- Nodes can fail by crashing or exhibiting Byzantine behavior
- Cryptographic primitives (digital signatures, hash functions) are secure

*2) Adversary Model:* We consider a computationally bounded adversary that can:

- Control up to $f$ out of $n$ nodes in any component
- Schedule message delivery within network assumptions
- Observe all network traffic (but not decrypt properly secured messages)
- Attempt to violate safety or liveness properties

The adversary cannot break cryptographic assumptions or control more than a threshold fraction of nodes in any component.

*3) Trust Model:* LIBR operates under a *trust-but-verify* model where:

- Moderators are assumed to act honestly but may be compromised
- Database nodes are assumed to be rational (not necessarily altruistic)
- Clients may be malicious but cannot forge signatures

## V. LIBR Architecture and Protocols

### A. System Architecture

LIBR employs a three-tier architecture designed for modularity and fault tolerance:

*1) Clients:* Clients serve as user interfaces and protocol coordinators. They are responsible for:

- Message composition and submission
- Signature collection from moderators
- Database node discovery and interaction
- Content retrieval and presentation

*2) Moderators:* Moderators form the governance layer, implementing community-driven content policies. Each moderator maintains:

- Cryptographic key pair for signature generation
- Community-specific moderation rules (though may choose arbitrary rules)
- Governance-specific metrics

*3) Database Nodes:* Database nodes provide persistent storage through a replicated DHT. They implement:

- Consistent hashing for data distribution
- Replication for fault tolerance
- Integrity verification for stored content

### B. Core Protocols

*1) Message Submission Protocol:* Algorithm 1 describes the message submission process:

**Algorithm 1** Message Submission Protocol

**procedure** SUBMITMESSAGE($m$, $community$)
  $ts \leftarrow getCurrentTime()$
  $sign \leftarrow sign(m, ts, user.pKey)$
  $moderators \leftarrow getModerators(community)$
  $modSigns \leftarrow \{\}$
  **for** $mod \in moderators$ **do**
    $response \leftarrow requestModSign(mod, m, ts, sign)$
    **if** $response.approved$ **then**
      $modSigns \leftarrow modSigns \cup \{response.sign\}$
    **end if**
  **end for**
  **if** $|modSigns| > \frac{|moderators|}{2}$ **then**
    $modCert \leftarrow createCertificate(m, ts, modSigns)$
    $dbNodes \leftarrow selectDbNodes(ts, community.R)$
    **for** $node \in dbNodes$ **do**
      $store(node, m, modCert)$
    **end for**
    **return** SUCCESS
  **else**
    **return** INSUFFICIENTAPPROVALS
  **end if**
**end procedure**

*2) Byzantine Consensus for Moderation:* Moderators employ a simple majority Byzantine consensus protocol. For a community with moderators $M = \{m_1, m_2, \ldots, m_n\}$, a message is approved if:

$$|\text{Approvers}| > \frac{|M|}{2} \tag{1}$$

This ensures that approval requires a simple majority of moderators, preventing minority censorship while enabling

efficient decision-making with equal voting power for all moderators.

*3) Database Node Selection:* Database nodes are selected using consistent hashing to ensure deterministic placement and even load distribution. The selection uses the FIND_NODE operation as offered in the libp2p Kademlia DHT Implementation

$$\text{primary_node} = FIND_NODE(hash(timestamp)) \quad (2)$$

Additional replica nodes are selected using integer nonce values, ensuring reproducible selection across all network participants.

$$\text{ith_replica} = FIND_NODE(hash(timestamp||i)) \quad (3)$$

### C. Partial Immutability Mechanism

LIBR implements partial immutability through cryptographic certificates and distributed storage:

1) **Integrity Protection**: Each message is cryptographically signed by its author and approved moderators.
2) **Distributed Storage**: Messages are replicated across multiple database nodes, preventing single points of failure.
3) **Audit Trail**: All moderation decisions are recorded and verifiable through cryptographic signatures.

## VI. Incentivization and Governance Model

The incentivization and governance mechanisms of our system are designed to ensure fair compensation for database nodes while maintaining a robust community-driven moderation structure. These mechanisms operate on a foundation of cryptographically secured smart contracts deployed on the Ethereum blockchain. Note that this section does not focus on the actual source of revenue generation, just using a part of it in governance and incentivization

### A. Incentivization Mechanism

The incentivization scheme ensures that database nodes (DB nodes) are rewarded based on their contribution to the network. The core components of the incentivization mechanism are as follows:

*1) Proof of Service for Database Nodes:* A Proof of Service (PoSe) is established to quantify the contributions of each DB node. This is achieved through the maintenance of a service score, computed and verified by moderators (acting as governors in the governance model). The process includes:

1) **Acknowledgements:** Each time a DB node stores a message, it sends an acknowledgment to the corresponding moderator.
2) **Random Queries:** Moderators randomly query stored messages to verify the DB node's service over time, initially with higher frequency and gradually decreasing.
3) **Score Calculation:** Moderators maintain and periodically update scores for each DB node based on acknowledgements and query results.

*2) Token Distribution:* At the end of each epoch (e.g., one month), tokens from a shared community revenue pool are distributed to DB nodes. The process is as follows:

1) **Score Submission:** Moderators submit scores for DB nodes to the blockchain.
2) **Transparency Window:** A public review period allows moderators and the community to audit scores for potential bias or inaccuracies.
3) **Challenge Mechanism:** If a score is disputed and a majority of moderators agree, the score is invalidated and the responsible moderator is penalized.
4) **Salary Collection:** DB nodes collect their tokens by submitting an **Aggregated Proof of Service Document**, cryptographically verifiable, which includes:
   - The initial date of service.
   - The total tokens received to date.

   The smart contract verifies this document, updates it with the new salary, and transfers the tokens to the DB node.

### B. Governance Model

The governance model integrates incentivization with moderator elections and oversight mechanisms to ensure balanced power distribution and fairness.

*1) Eligibility for Voter and Moderatorship:* A DB node qualifies as a valid elector for moderation elections after providing a minimum PoSe threshold (e.g., three epochs of service and a specific number of tokens earned).

Any potential voter can contest for the elections by staking a fixed amount of tokens on the chain

*2) Election Process:* Elections for moderators occur at the end of each epoch. The process includes public voting by the valid electors and previous moderators (excluding those penalized to vote).

*This scheme relies on public voting, as anonymous voting might be expensive and non-scalable for the use-case. Though alternatives are being explored*

To vote, a valid elector need not interact with the chain previously, but in just one transaction, providing its vote along with the Proof of Service.

The elected moderators (or governors) are stored on-chain for already discussed reasons.

*3) Moderator Oversight:* Mechanisms are in place to prevent arbitrary actions by moderators and address moderation gaps:

- **Bidirectional Ticketing System:** Clients can raise tickets for two types of moderation issues:
  - *Over-moderation tickets*: For content they believe was wrongly censored
  - *Under-moderation tickets*: For content that should have been censored but remains visible in the forum

  The raised ticket is broadcasted to the moderators, who each return a signed message determining the ticket's "actionability"

- **Consensus Verification:** If a majority of moderators find the ticket actionable, the client can escalate the ticket on-chain
- **Retroactive Moderation:** For under-moderation tickets, if consensus is reached that content should be removed, moderators can issue a removal certificate that instructs database nodes to mark the content as removed
- **Penalties:** Moderators found guilty of biased actions or systematic moderation failures are penalized, including loss of tokens and disqualification from future elections

### C. Integration of Incentivization and Governance

The incentivization and governance systems are interconnected:

- DB nodes are incentivized to maintain high service quality to improve their eligibility for moderatorship.
- Moderators ensure fairness in scoring and salary distribution, which directly impacts DB nodes' reputation and rewards.

This dual-layered approach ensures a balanced, self-regulating ecosystem that aligns individual incentives with community goals.

## VII. Security Analysis

### A. Formal Properties

We prove that LIBR satisfies the following security properties under standard Byzantine assumptions:

*1) Safety:* **Theorem 1**: No two conflicting moderation decisions can be simultaneously valid.

**Proof Sketch**: Suppose, for contradiction, that a message $m$ receives both approval and rejection certificates at time $t$. This requires majority approval from moderators for both decisions simultaneously. The sum of approvals and disapprovals excede the total number of moderators, as a single moderator can either approve or disapprove for the same message, thus it is impossible. □

*2) Liveness:* **Theorem 2**: Every message submitted by an honest client will eventually be processed by honest moderators.

**Proof Sketch**: Under partial synchrony assumptions, messages eventually reach all honest moderators. Honest moderators process messages according to protocol rules, and approval requires only a majority of honest moderators. Since Byzantine moderators constitute less than half the total weight, honest moderators can always reach consensus. □

*3) Censorship Resistance:* **Theorem 3**: Messages approved by honest moderators cannot be permanently censored by Byzantine actors.

**Proof Sketch**: Approved messages are stored on multiple database nodes selected through deterministic hashing. Byzantine database nodes cannot prevent storage on honest nodes, and the replication factor ensures availability as long as at least one honest replica exists. □

### B. Attack Analysis

*1) Sybil Attacks:* LIBR mitigates Sybil attacks through cryptographic identity binding and community-based moderator selection. Moderators must be explicitly approved by the community, preventing arbitrary identity creation.

*2) Eclipse Attacks:* The deterministic database node selection prevents eclipse attacks by ensuring content is stored on predictable nodes that cannot be controlled by the adversary without violating Byzantine assumptions.

*3) Availability Attacks:* With replication factor $R$ and $n$ database nodes, where each node has availability probability $p$, the probability that content remains available is:

$$P(\text{available}) = 1 - (1-p)^R \tag{4}$$

For typical values ($R = 3$, $p = 0.9$), this yields 99.9% availability.

## VIII. Theoretical Analysis and Hypothesis Validation

Since LIBR represents a novel framework design, we present comprehensive theoretical analysis and simulation-based validation of our design principles rather than experimental evaluation of a full implementation.

### A. Research Hypotheses

We formulate and analyze the following core hypotheses:

**H1 (Performance Hypothesis)**: LIBR achieves $O(\log n)$ message complexity for content operations while maintaining sub-second latency for content retrieval.

**H2 (Fault Tolerance Hypothesis)**: LIBR maintains system availability $> 99\%$ with up to $\lfloor \frac{m-1}{2} \rfloor$ Byzantine moderators and up to $\frac{n}{3}$ Byzantine database nodes.

**H3 (Scalability Hypothesis)**: LIBR scales logarithmically with network size while maintaining consistent performance characteristics.

**H4 (Security Hypothesis)**: LIBR prevents censorship attacks and maintains content integrity under standard Byzantine assumptions.

### B. Theoretical Performance Analysis

*1) Complexity Analysis:* **Message Submission Complexity**:

- Moderator consultation: $O(m)$ where $m$ is number of moderators
- Database node selection: $O(\log n)$ using DHT lookup
- Storage operations: $O(R)$ where $R$ is replication factor
- Total complexity: $O(m + \log n + R)$

**Content Retrieval Complexity**:

- DHT lookup: $O(\log n)$
- Certificate verification: $O(m)$ for signature verification
- Total: $O(\log n + m)$

For practical deployments where $m \ll n$, this simplifies to $O(\log n)$, validating H1.

*2) Latency Analysis:* Expected latency for message submission:

$$E[L_{submit}] = L_{network} + L_{crypto} + L_{consensus} + L_{storage}$$

Where:

- $L_{network} = O(\Delta)$ (network delay)
- $L_{crypto} = O(m \cdot T_{sig})$ (signature operations)
- $L_{consensus} = O(\Delta)$ (consensus delay)
- $L_{storage} = O(R \cdot \Delta)$ (storage replication)

For typical parameters ($m = 5$, $R = 3$, $\Delta = 100ms$, $T_{sig} = 1ms$):

$$E[L_{submit}] \approx 100 + 5 + 100 + 300 = 505ms$$

This validates our sub-second latency claim in H1.

### C. Availability Analysis

*1) Moderator Availability:* With $m$ moderators and threshold $t = \lceil \frac{m+1}{2} \rceil$, system availability is:

$$A_{mod} = \sum_{k=t}^{m} \binom{m}{k} p^k (1-p)^{m-k}$$

Where $p$ is individual moderator availability.

For $m = 5$, $t = 3$, $p = 0.9$:

$$A_{mod} = \binom{5}{3}(0.9)^3(0.1)^2 + \binom{5}{4}(0.9)^4(0.1)^1 + \binom{5}{5}(0.9)^5 \tag{5}$$

$$= 10 \times 0.729 \times 0.01 + 5 \times 0.6561 \times 0.1 + 1 \times 0.59049 \tag{6}$$

$$= 0.0729 + 0.32805 + 0.59049 = 0.99144 \tag{7}$$

*2) Storage Availability:* With $R$ replicas and node availability $p$:

$$A_{storage} = 1 - (1-p)^R$$

For $R = 3$, $p = 0.95$:

$$A_{storage} = 1 - (0.05)^3 = 0.999875$$

Overall system availability: $A_{system} = A_{mod} \times A_{storage} = 0.991$

This validates H2 with $> 99\%$ availability.

### D. Comparison with Existing Systems

Table I compares LIBR with existing social media platforms across key governance and architectural dimensions:

LIBR uniquely combines high censorship resistance with effective community-driven moderation through Byzantine fault-tolerant governance, addressing limitations of both centralized and existing decentralized approaches.

### E. Theoretical Complexity Comparison

Table II provides a detailed comparison of theoretical properties:

LIBR achieves optimal latency bounds while maintaining strong Byzantine tolerance and censorship resistance.

TABLE I
PLATFORM COMPARISON

| Platform | Censorship Resistance | Moderation Quality | Governance Model | Incentivization Mechanism |
|---|---|---|---|---|
| *Centralized Platforms* | | | | |
| Reddit | None | High | Corporate | - |
| Twitter/X | None | Medium | Corporate | - |
| *Decentralized Platforms* | | | | |
| Steemit | Full | Low | Token-weighted | Economic Rewards |
| Mastodon | Partial | Medium | Instance-based | Voluntary |
| Nostr | Full | None | Relay-based | Volunteer (Lightning Tips) |
| IPFS Forums | Full | None | None | None |
| **LIBR** | **Partial** | **High** | **Community BFT** | **Service-based** |

TABLE II
THEORETICAL COMPLEXITY COMPARISON

| System | Consensus Complexity | Storage Complexity | Latency Bound | Byzantine Tolerance | Censorship Resistance |
|---|---|---|---|---|---|
| Blockchain | $O(n)$ | $O(n)$ | $O(block_time)$ | $< 33\%$ | High |
| Federated | $O(1)$ | $O(1)$ | $O(\Delta)$ | $0\%$ | Low |
| Pure P2P | $O(\log n)$ | $O(\log n)$ | $O(\log n \cdot \Delta)$ | $< 50\%$ | High |
| **LIBR** | $O(m)$ | $O(\log n)$ | $O(\Delta)$ | $< 50\%$ | High |

## IX. DISCUSSIONS AND FUTURE WORK

### A. Discussions

The current implementation of LIBR demonstrates the feasibility of a censorship-resistant social network framework that balances decentralization with effective moderation. The incentivization scheme enhances economic viability, addressing a critical gap in sustainable operations for decentralized platforms. This approach ensures fair compensation for database nodes and encourages active participation in governance.

The inclusion of advanced moderation techniques, such as leveraging Google Cloud NLP, improves content quality while maintaining efficiency. By utilizing cryptoeconomic principles and community-driven moderation, LIBR effectively bridges the gap between censorship resistance and usability.

Additionally, the development of mobile and web client interfaces aims to significantly enhance user accessibility, thereby broadening adoption potential. These efforts underscore the practicality of LIBR in real-world applications.

### B. Future Work

Several areas warrant further enhancement:

- **Enhanced Scalability:** While LIBR scales logarithmically with network size, the consensus process for moderation introduces bottlenecks for extremely large communities. Future iterations could explore hierarchical moderation or sharding approaches to address these limitations.
- **Privacy Improvements:** The current pseudonymous design could benefit from stronger privacy guarantees, such as integrating zero-knowledge proofs or mix networks, to protect user identities while preserving accountability.
- **Cross-Platform Interoperability:** Establishing standardized protocols to enable federation with other decentralized platforms would foster broader adoption and enhance LIBR's ecosystem.
- **User Experience:** The development of fully functional mobile and web clients remains a priority, with a focus on

intuitive interfaces and seamless interactions to increase user engagement and adoption.

- **Experimental Validation:** A comprehensive evaluation of LIBR under real-world conditions, including user studies and large-scale simulations, would provide valuable insights into performance, usability, and scalability.

By addressing these challenges, LIBR can evolve into a more robust and versatile framework, contributing significantly to the ongoing development of decentralized communication platforms.

## X. Conclusion

In this work, we have presented LIBR, a novel framework for censorship-resilient social networking that unifies decentralized storage with community-driven moderation. By leveraging Byzantine fault-tolerant consensus among cryptographically-verified moderators, LIBR achieves a unique balance between free expression and content quality assurance. Our formal security analysis demonstrates that LIBR satisfies safety, liveness, and censorship-resistance guarantees under standard Byzantine assumptions. Theoretical complexity bounds further show that key operations—message submission, retrieval, and moderation—scale logarithmically with network size, enabling practical deployment in large communities.

To ensure economic sustainability, we introduced a cryptoeconomic incentivization scheme that rewards database nodes based on verifiable Proof-of-Service, fostering reliable participation without central coordination. Advanced moderation support, including integration with Google Cloud NLP, enhances content analysis while preserving human oversight. Finally, the forthcoming mobile and web clients promise to broaden accessibility and drive real-world adoption of LIBR's principles.

Looking ahead, LIBR's open-source implementation provides a solid foundation for further enhancements—ranging from hierarchical moderation and privacy-preserving zero-knowledge techniques to cross-platform federation and large-scale user studies. We believe that LIBR charts a viable path toward next-generation digital forums that uphold both community standards and censorship resistance, and we invite the research community to build upon and extend its capabilities.

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